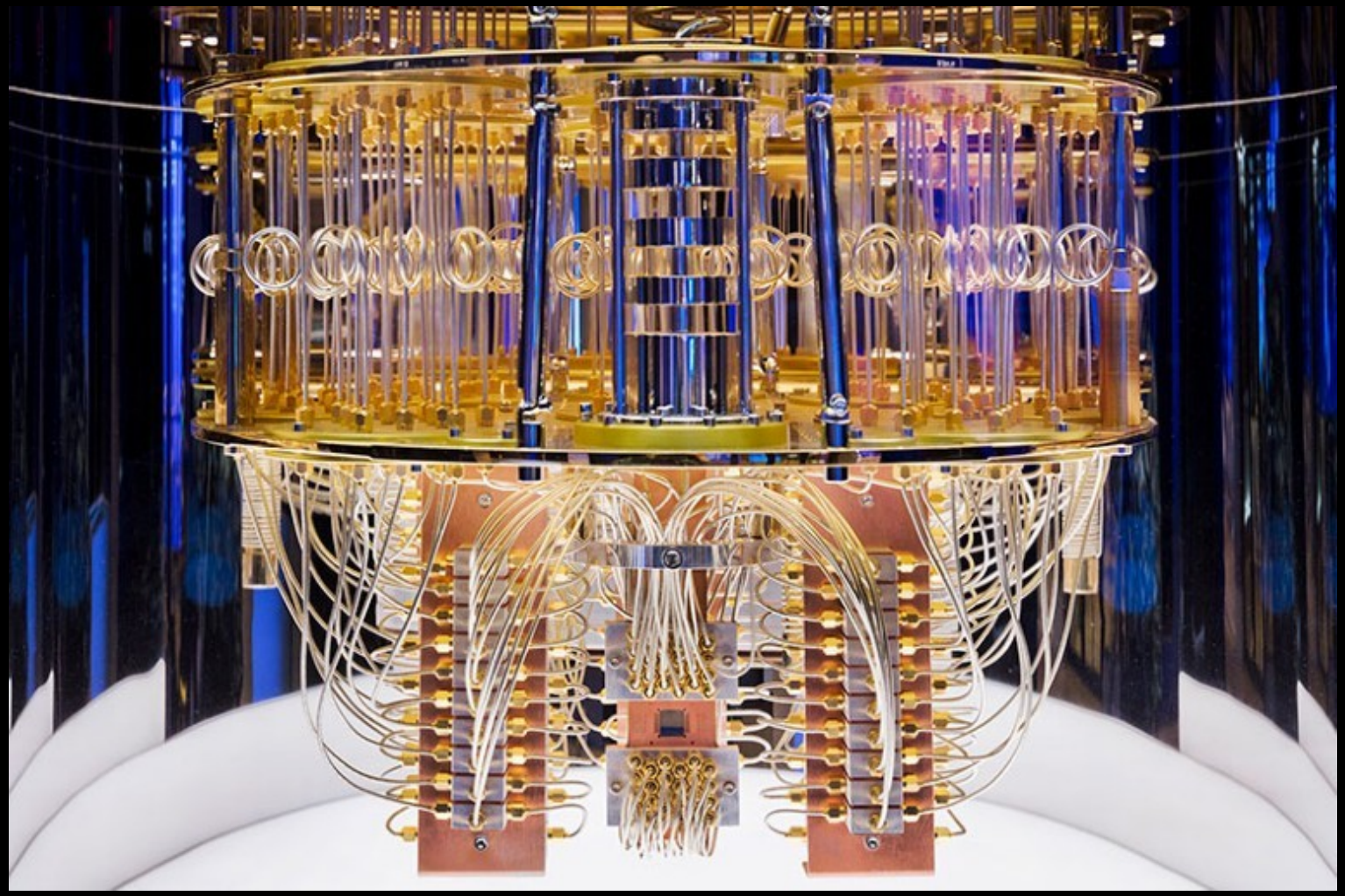
The Quantum Threat to Bitcoin



bitcoinbushbash.info

Is Bitcoin fucked?

No*

What’s the threat → How we fix it → What you can do

What is quantum computing?

Quantum mechanical phenomena

Superposition: many states at once

Entanglement: correlated states

Interference: amplify right answers

How they work

Qubit: 0 and 1 at the same time

Circuit: superposition & entanglement gates

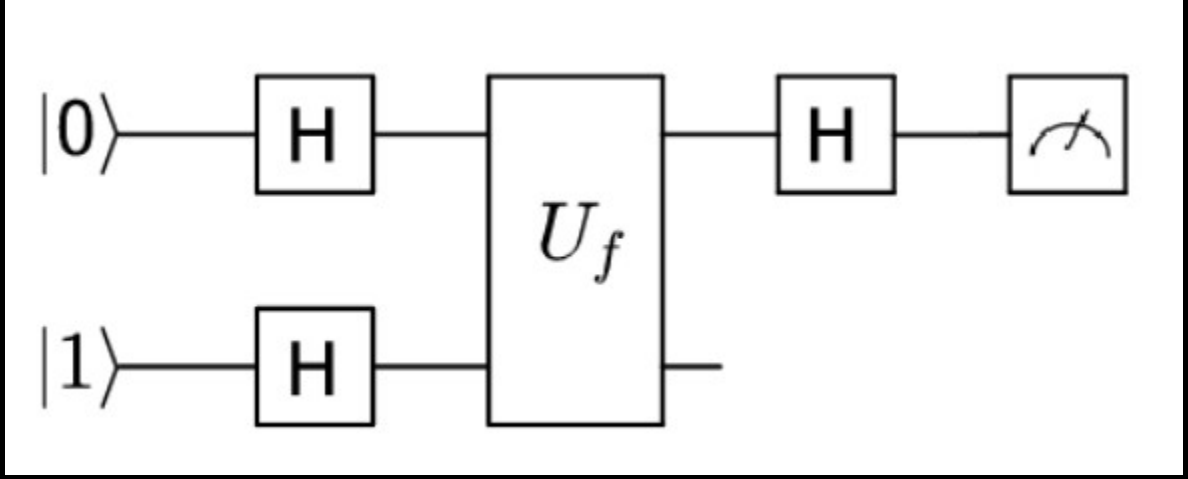
Measure: interference amplifies right answer

Fault tolerance

Qubits are fragile and can’t be copied

Physical qubits vs Logical qubits

CRQC requires ~2300 logical qubits



The QC threat to Bitcoin

Shor's algorithm (signatures)

- Breaks ECC and RSA
- Derives private key from public key
- Exponential speedup

Grover's algorithm (hashing)

- Targets SHA-256 / RIPEMD-160
- Quadratic speedup only
- 256-bit -> ~128-bit effective
- Quantum mining favors centralisation

Two types of QC attack

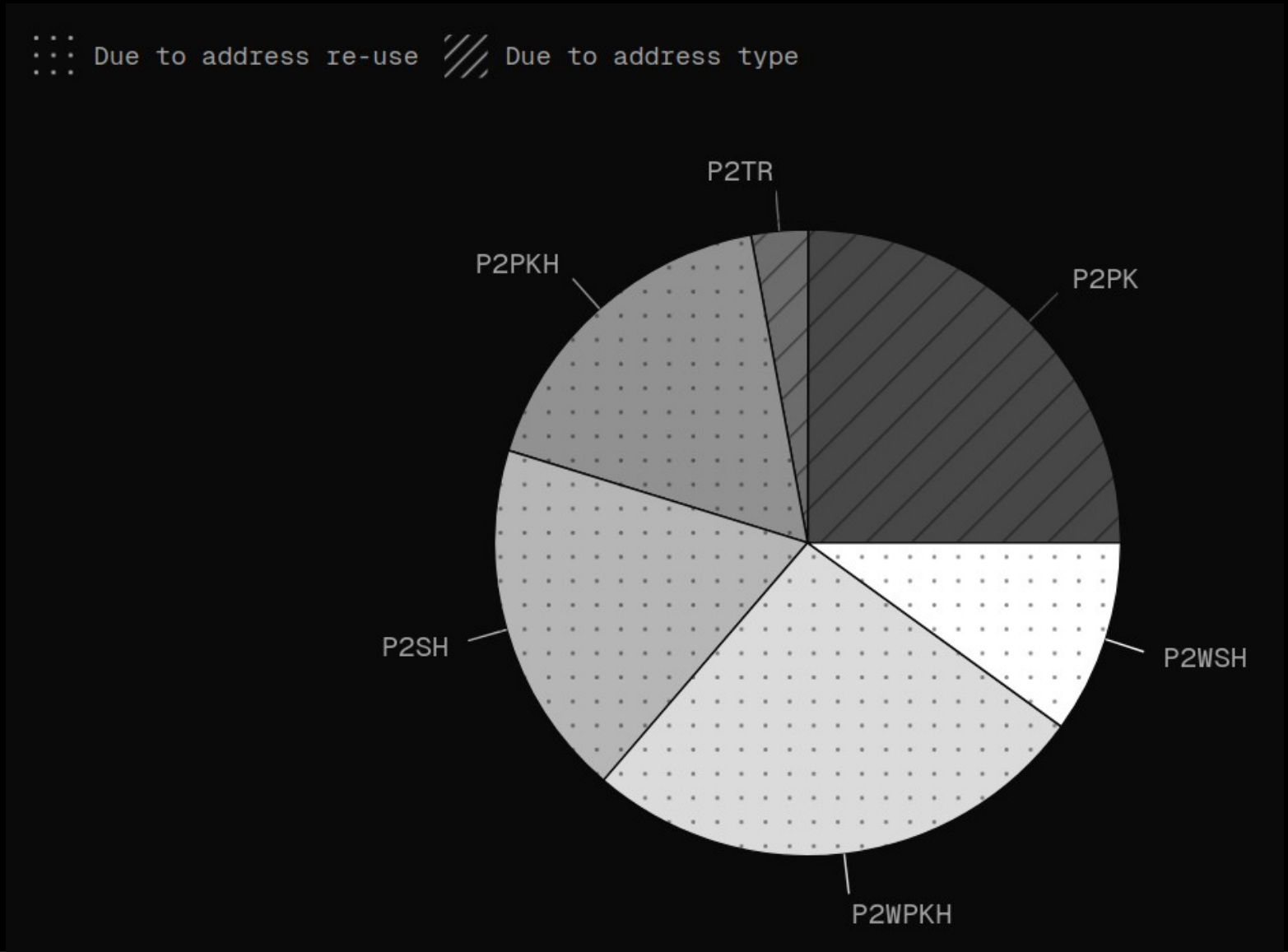
Long exposure

Short exposure

Long exposure attack vulnerability

Type	Vulnerable	Prefix	Example
P2PK	Yes	Varies	02103203b768951584fe9af6d9d9e6ff26a5f76e453212f19ba163774182ab8057f3eac
P2PKH	No*	1	1A1zP1eP5QGefi2DMPTfTL5SLmv7DivfNa
P2MS	Yes	Varies	52410496ec45f878b62c46c4be8e336dff7cc58df9b502178cc240e...
P2SH	No*	3	3FkhZo7sGNue153xhgqPBcUaBsYvJW6tTx
P2WPKH	No*	bc1q	bc1qsnh5ktku9ztqeqfr89yrqjd05eh58nah884mku
P2WSH	No*	bc1q	bc1qvhu3557twysq2ldn6dut6rmaj3qk04p60h9l79wk4lzgy0ca8mfsnffz65
P2TR	Yes	bc1p	bc1p92aslsnseq786wxfk3ekra90ds9ku47qttupfjsqmmj4z82xdq4q3rr58u
P2MR	No*	bc1z	bc1zzmv50jjgxxhww6ve4g5zpewrkjqhr06fyujpm20tuezdlxmfphcqfc80ve

Long exposure attack vulnerability



6,874,398 BTC

When is it coming?

MARCH 2026

QUANTUM THREAT TIMELINE REPORT 2025

Authors
Dr. Michele Mosca, Co-Founder & CEO, evolutionQ Inc.
Dr. Marco Piani, Senior Research Analyst, evolutionQ Inc.

When is it coming?

GRI Quantum Threat Timeline Report 2025 — 26 experts surveyed

~15%

Optimistic avg within 5 years

~49%

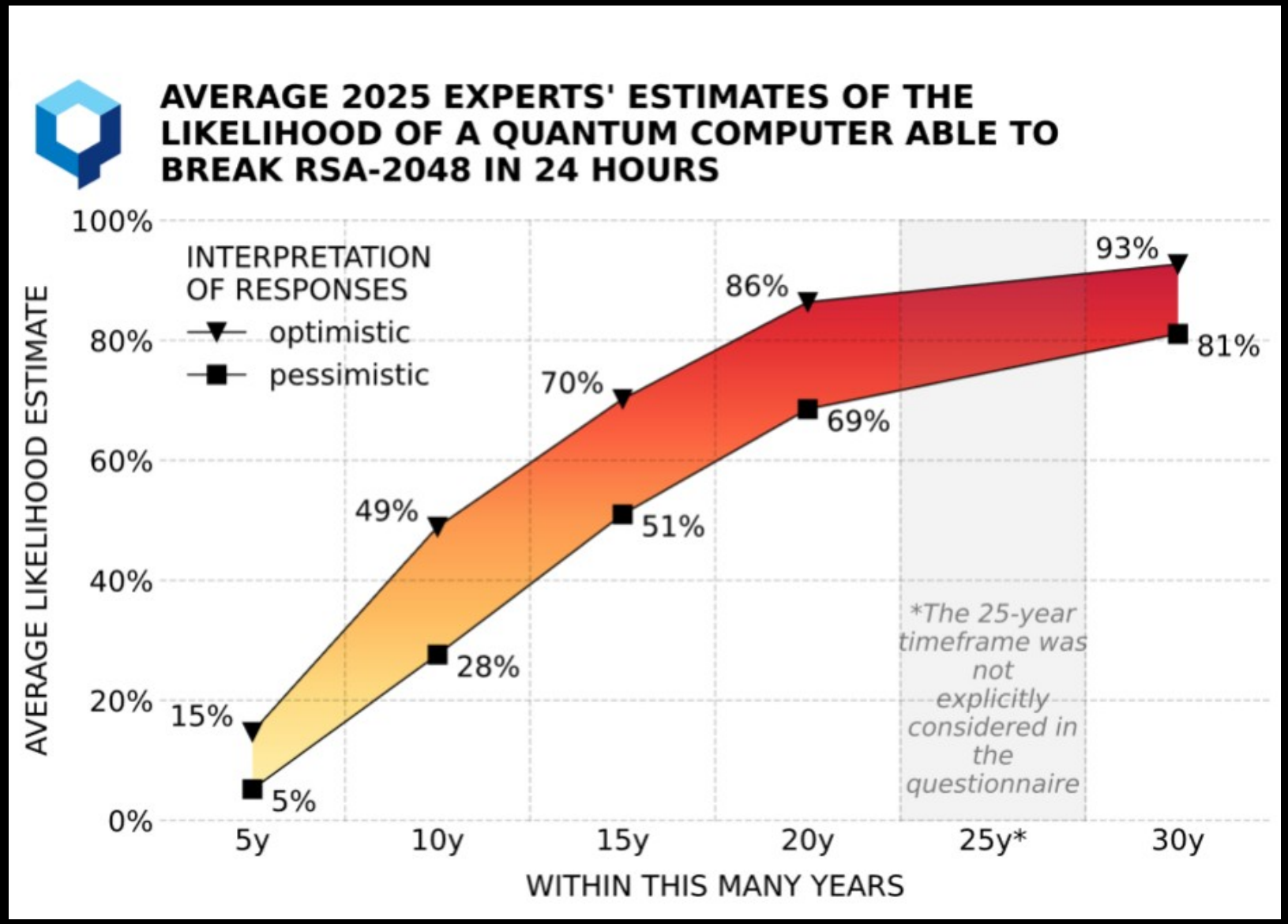
Optimistic avg within 10 years

~28%

Pessimistic avg within 10 years

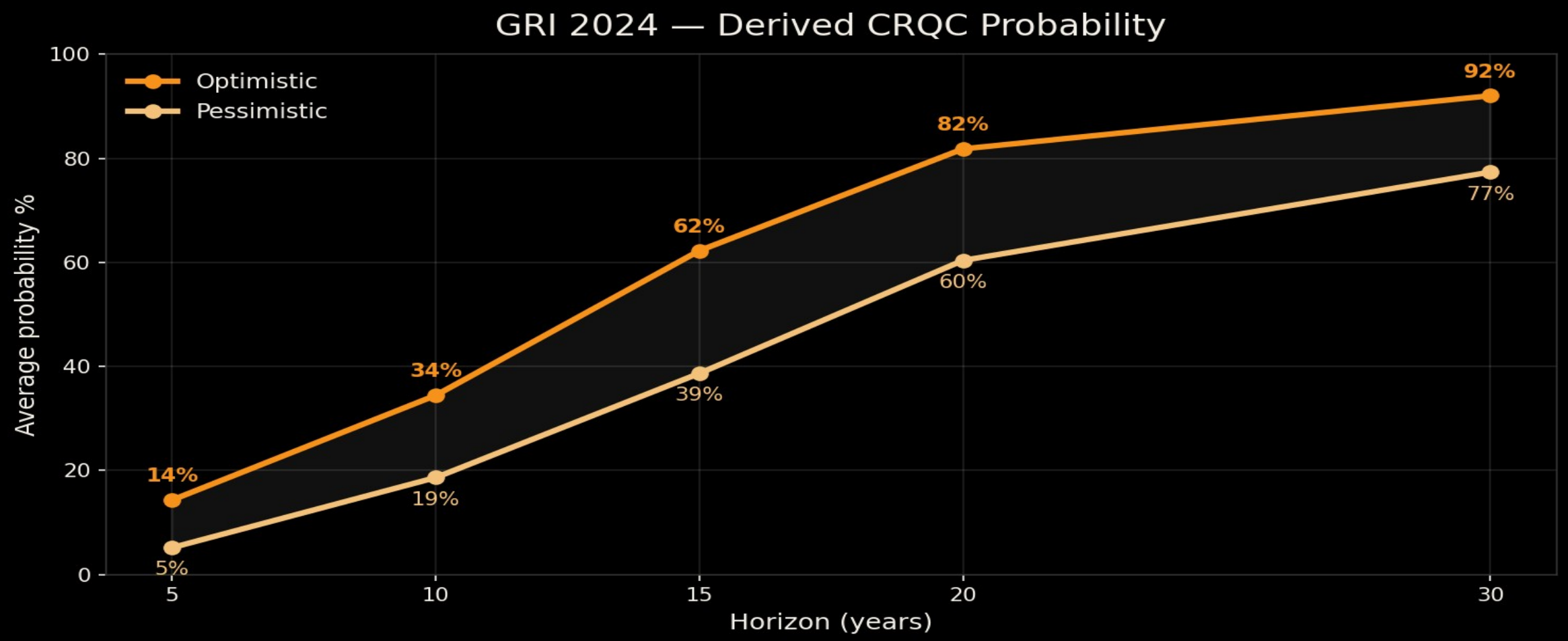
Also look at Bitcoin Fundamentals 11 Nov 25 podcast with Charles Edwards

When is it coming?



When is it coming?

Optimistic vs pessimistic probability assignments



Government policy timelines

2030

CNSA 2.0
PQC mandate begins

2033

CNSA 2.0
full browser/OS PQC

2035

NIST IR 8547
disallow ECC

Bitcoin doesn't answer to regulators — but enterprise/institutional users will demand PQC.

Bitcoin Revision

Public key cryptography: ECC, signature algorithms

Hashing: mining, address compression, check sums, key derivation, Merkle trees

Public key hidden by hash until first spend (P2PKH/P2SH/P2WSH)

Public key exposed on receive (P2PK, P2TR key-path)

Mempool window: tx waits ~10 min before confirmation

When your public key is visible = your quantum exposure level

The two big Bitcoin problems

1. Technical: add post-quantum cryptography to Bitcoin
2. Social: what happens to legacy coins that don't upgrade

Post-quantum signature options

NIST standardised

ML-DSA (Dilithium) — lattice-based

SLH-DSA (SPHINCS+) — hash-based

Conservative, well-studied

Compact sigs, larger public keys

Other families

Code-based (McEliece) — huge keys

Multivariate (Rainbow — broken)

Isogeny (SIKE — broken)

Active research, real risks

Signature Comparisons

	ECDSA (secp256k1)	ML-DSA-44	SLH-DSA-128s
Public Key	33 B	1,312 B	32 B
Private Key	32 B	2,560 B	64 B
Signature Size	71 B	2,420 B	7,856 B
Signing Time	~0.01 ms	~0.24 ms	~94 ms
Verification Time	~0.01 ms	~0.05 ms	~6 ms
Quantum Safe	No	Yes	Yes
Based On	Elliptic curves	Lattices	Hash functions

BIP-360: Pay-to-Merkle-Root

P2MR

New output type
SegWit version 2

bc1z...

New address
prefix (Bech32m)

Soft fork

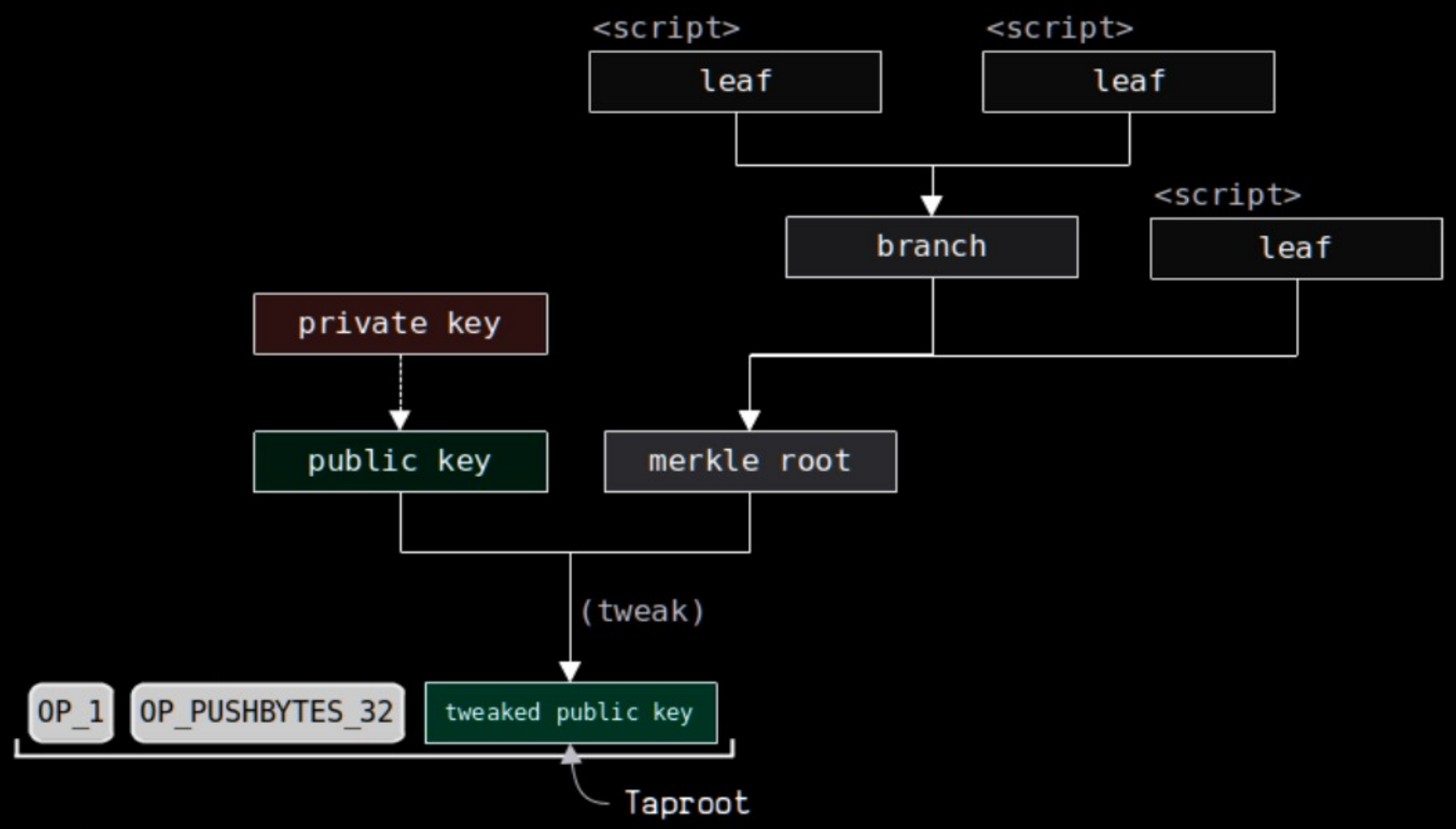
Backward
compatible

Removes key-path spend — no public key exposed on-chain

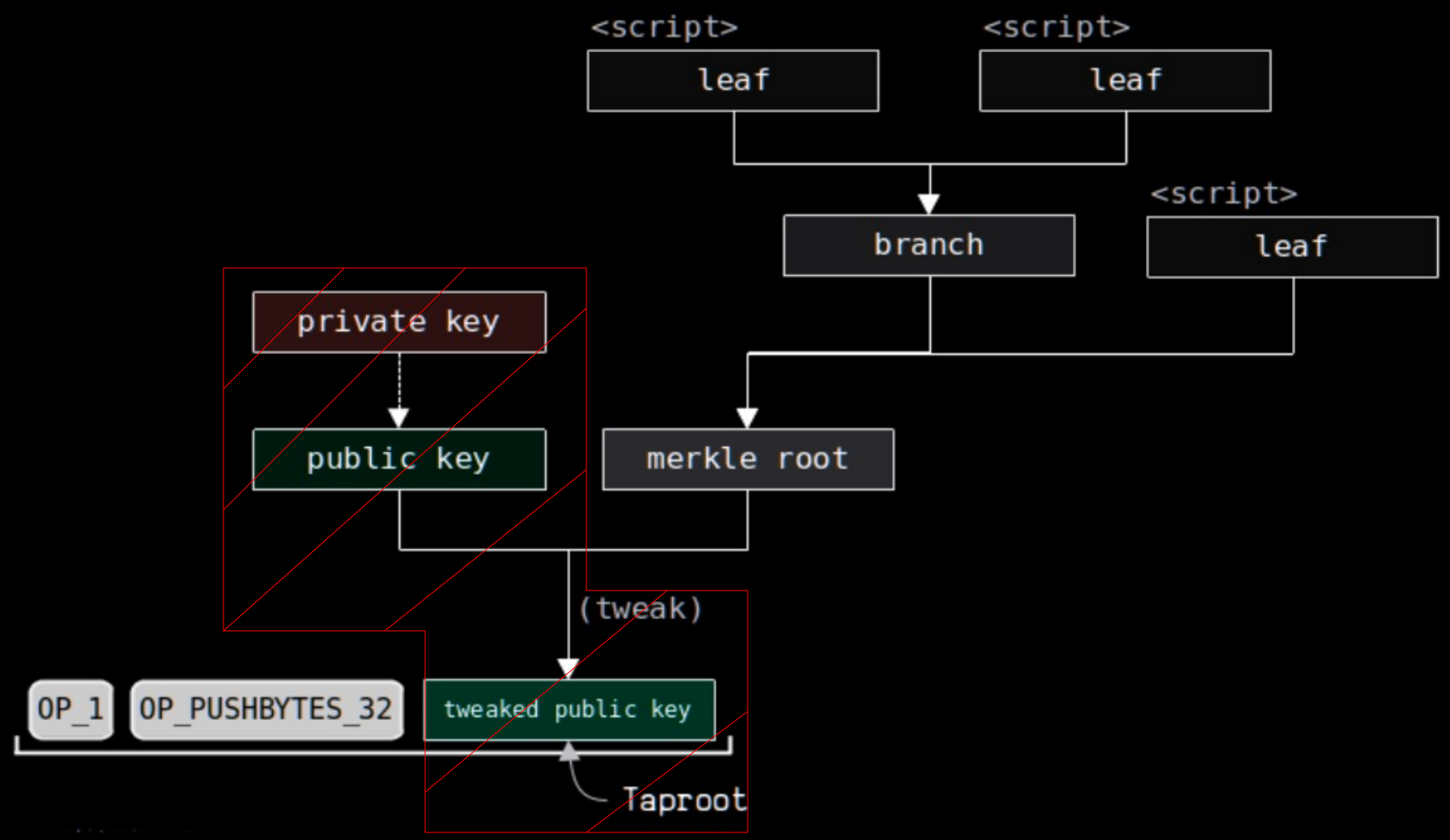
Reuses battle-tested tapscript (BIPs 340/341/342)

Creates safe upgrade path for future PQC opcodes

P2TR Structure



P2MR Structure



BIP-360: design principles

Rationale

Minimizes changes to the network

Supports tapscript which assists in implementing PQC required opcodes

Facilitates gradual integration of PQC without negative impacts before Q-Day.

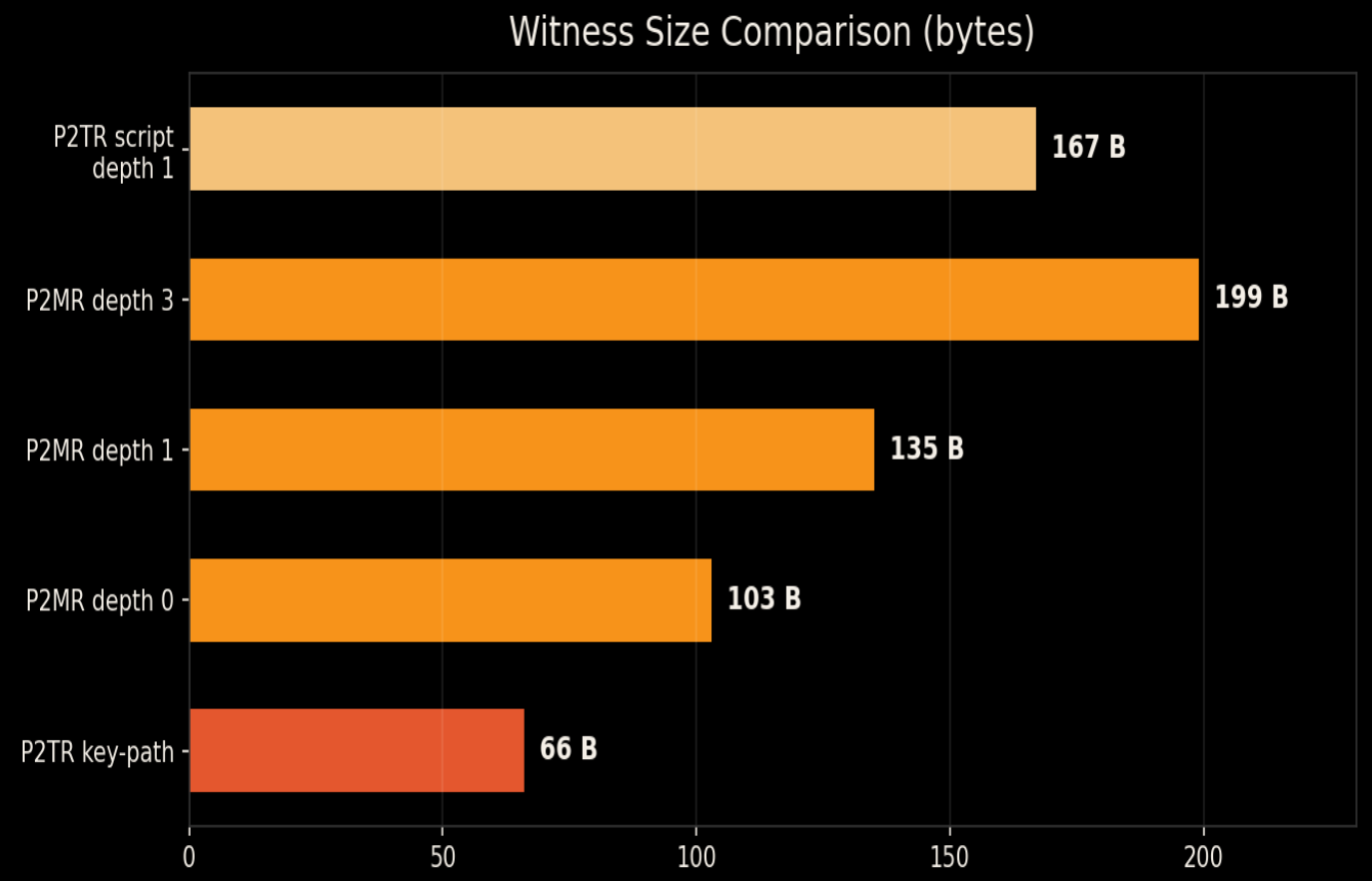
Trade offs

Size: larger than key path spend, smaller than script path spend

Privacy: can’t hide you have other script path spends (compared to key path spend)

BIP-360: witness size tradeoffs

Larger than P2TR key-path — smaller than P2TR script-path



Depth 0 (single leaf): 103 B

vs P2TR key-path: 66 B

+37 B base overhead

+32 B per tree level

Always 32 B smaller than equivalent P2TR script-path (no internal key in control block)

BIP-360: what it protects

Protected: long exposure

Public keys hidden in the Merkle tree

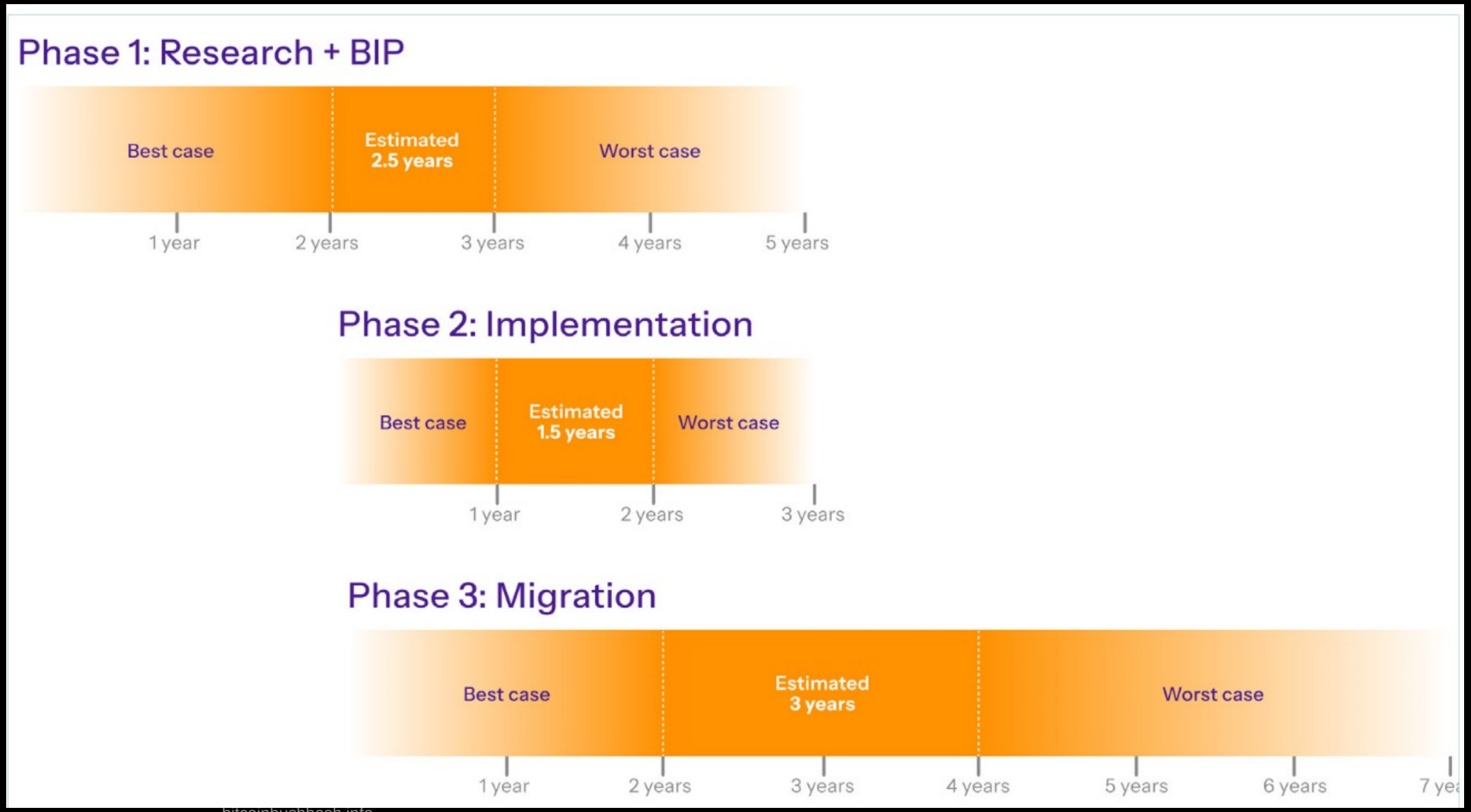
Not yet protected: short exposure

Mempool window still uses ECDSA/Schnorr

Needs future PQC signature opcodes

P2MR creates the path for these

Implementation timeline



Problem 2: legacy coins

Options

Status quo: let CRQC spend them, preserve neutrality

Freeze: lock vulnerable UTXOs preemptively

Hourglass: rate-limit movement of exposed coins

Consequences

Steal: price shock, but censorship resistance intact

Freeze: market crash prevented, legitimacy damaged

Hourglass: middle path, still contentious

What can you do?

Today

Don't reuse addresses

Avoid P2TR for large holdings

Don't share your XPUB unnecessarily
(SMSF auditing, public Electrum nodes)

Satoshi's shield: game-theory deterrent

When standards land

Migrate when upgrades are made

Adopt tested PQC implementations

Run a node to support the upgrade

Fund dev grants and crypto research

Participate in the consensus direction

Mythbusters

Myth	Reality
CRQC breaks all addresses instantly	Exposure type and time window matter
Grover kills SHA-256	Quadratic speedup shifts cost curves, not instant break
This is only a Bitcoin problem	ECDSA/RSA break is an internet-wide crisis

CRQC most likely use cases

SIGINT: crack the 'store now, decrypt later' vault

Materials science and engineering simulation

Pharmaceutical drug discovery

Disease prevention and genomics research

Physics: push the boundary of QFT

Questions?

Priority 1: stop exposing public keys

Priority 2: safe upgrade path (BIP-360)

Priority 3: social consensus on legacy coins

References (1/10)

What is QC

1. IBM Quantum Learning: Bits, Gates, and Circuits

IBM's technical introduction to qubits, quantum gates, and circuits — aimed at people with a computing background, no physics degree required.

2. Classiq: The Deutsch-Jozsa Algorithm Explained

Step-by-step walkthrough of the Deutsch oracle problem and circuit — the simplest demonstration that a quantum computer can outperform a classical one.

3. Bitcoin Magazine: Quantum Bitcoin Summit

Summary of a summit bringing together quantum computing and Bitcoin experts to examine the real threats and separate hype from substance.

References (2/10)

What is the threat

4. Grover's Search Algorithm Overview

Microsoft's explanation of Grover's quantum search algorithm, which provides a quadratic speedup for unstructured search problems.

5. Quantum Resource Estimates for ECDLP

Roetteler et al. (2017) foundational paper estimating ~2,330 logical qubits and ~1.29×10^{11} Toffoli gates to break 256-bit ECC using Shor's algorithm.

6. Ha, Lee & Heo (2024) — Resource analysis for ECDLP with noisy qubits

First comprehensive physical qubit estimate for ECDLP under surface codes: ~5.87 million physical qubits for P-256 at 10^{-3} error rate with 200 ns cycle time.

7. Ed25519 Optimisation (2025)

Achieves 97% reduction in T-count, 60% reduction in T-depth, and 16% fewer qubits for quantum attacks on Ed25519 by exploiting its specific finite field structure.

References (3/10)

Bitcoin revision

8. Elliptic Curve Cryptography
How Bitcoin uses the secp256k1 elliptic curve for key generation and digital signatures — the operation Shor's algorithm can reverse.

9. Digital Signatures
How ECDSA and Schnorr signatures work in Bitcoin to authorise spending without revealing the private key.

10. Hash Functions
Overview of SHA-256 and RIPEMD-160 as used in Bitcoin for mining, addresses, and data integrity — Grover's algorithm provides only a quadratic speedup against these.

11. Memory Pool
The mempool where unconfirmed transactions wait — the window during which a short exposure quantum attack could occur.

12. Taproot (P2TR)
Pay-to-Taproot: exposes a tweaked public key on-chain, making unspent P2TR outputs vulnerable to long exposure quantum attacks.

References (4/10)

QC timeline

13. GRI Quantum Threat Timeline Report 2025

Annual survey of 26 global quantum experts. Optimistic interpretation: ~49% chance of a CRQC within 10 years; pessimistic: ~28%. Notable upward shift from prior years.

14. Bitcoin Fundamentals: Quantum Computing and Bitcoin

Podcast episode discussing the quantum threat to Bitcoin with Charles Edwards, covering timelines, risk assessment, and what bitcoiners should know.

15. Cointelegraph: BIP-360 Upgrade Timeline

BIP-360 co-author estimates Bitcoin may take up to 7 years to fully upgrade to post-quantum signatures under a standard timeline.

References (5/10)

Technical problem and fixes

16. BIP360.org Overview

Overview of Pay-to-Merkle-Root (P2MR), a new output type that removes the quantum-vulnerable key path spend from Taproot while preserving script tree functionality.

17. BIP-360 Draft: Pay-to-Merkle-Root

The formal BIP specification for P2MR, detailing address format (bc1z...), witness structure, and security rationale against long exposure quantum attacks.

18. BitMEX Research: Taproot Quantum Spend Paths

Proposes dual-path Taproot wallets with both quantum-safe and classical spend paths, arguing these wallets should be exempt from any future coin freeze.

19. Cointelegraph: Six Quantum Security Challenges

Outlines six major hurdles Bitcoin faces in becoming quantum-resistant, including signature size bloat, consensus changes, and migration logistics.

20. Bitcoin Magazine: Bitcoin Advances Toward Quantum Resistance

Reports on recent development progress toward quantum-resistant Bitcoin, including BIP-360 and related proposals.

References (6/10)

21. NIST FIPS 204: ML-DSA
NIST's finalised standard for ML-DSA (derived from CRYSTALS-Dilithium), a lattice-based post-quantum digital signature algorithm.

22. NIST FIPS 205: SLH-DSA
NIST's finalised standard for SLH-DSA (derived from SPHINCS+), a stateless hash-based post-quantum digital signature algorithm.

23. Post-Quantum Digital Signatures Benchmark: ML-DSA vs ECDSA
Side-by-side performance benchmarks comparing ML-DSA signature sizes, signing times, and verification times against classical ECDSA.

24. ASecuritySite: Digital Signature Benchmark
Interactive benchmarking tool for post-quantum signature schemes including Dilithium/ML-DSA across different security levels.

25. Making SLH-DSA 10x-100x Faster
Research by SLH-DSA co-designer Andreas Huelsing on optimisation techniques to dramatically improve SLH-DSA signing and verification performance.

References (7/10)

26. SLotH GitHub: SLH-DSA Performance
Reference implementation and benchmarks for optimised SLH-DSA, targeting hardware acceleration to close the performance gap with lattice-based schemes.

27. Chaincode: Bitcoin Post-Quantum
Chaincode Labs research paper analysing post-quantum migration strategies for Bitcoin, including signature scheme selection and upgrade paths.

References (8/10)

Social problem and fixes

28. Hourglass v2 Draft

Proposes a middle-ground approach to quantum-vulnerable coins: rate-limiting their movement rather than freezing or ignoring them entirely.

29. Lopp: Against Quantum Recovery of Bitcoin

Jameson Lopp argues Bitcoin should permanently burn funds in quantum-vulnerable addresses rather than allow quantum-capable entities to steal them.

30. CoinDesk: To Freeze or Not to Freeze — Satoshi and the $440B in Bitcoin Threatened by Quantum Computing

Examines the debate over ~7 million BTC in quantum-vulnerable addresses including Satoshi's ~1 million coins, weighing immutability against market stability.

31. Cointelegraph Magazine: Bitcoin may face hard fork over any attempt to freeze Satoshi's coins

Argues that freezing quantum-vulnerable coins would violate Bitcoin's core property rights, potentially splitting the network in a contentious hard fork.

References (9/10)

Other uses of QC

32. Frontiers: Quantum Computing — Foundations, Algorithms, and Emerging Applications (2025)

Comprehensive review covering quantum algorithms and their emerging applications across materials science, chemistry, optimisation, and physics simulation.

33. Wikipedia: Harvest Now, Decrypt Later

Overview of the HNDL strategy where adversaries intercept and store encrypted communications today for decryption once quantum computers become available.

References (10/10)

34. Federal Reserve: HNDL and Post-Quantum Cryptography Risks for Distributed Ledger Networks

Federal Reserve research examining how harvest-now-decrypt-later attacks pose data privacy risks specifically for blockchain and distributed ledger networks.

35. McKinsey: The Quantum Revolution in Pharma (2025)

McKinsey analysis of how quantum computing is accelerating drug discovery by simulating molecular interactions at the quantum level, compressing timelines from years to weeks.

36. Nature: Quantum-Machine-Assisted Drug Discovery (2025)

Nature paper on combining quantum computing with machine learning for biomarker discovery, genomic data processing, and cancer detection via liquid biopsy.